

Fig. 6: Actual-size PCB layout of colour identification

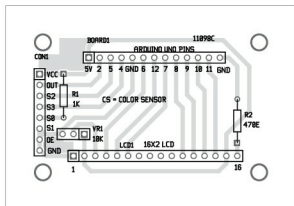


Fig. 7: Components layout of the PCB

Simulator IDE. The software is helpful for developing the project.

After the coding (written in BASIC) is done, compile the code or load the hex file generated into the ATmega328P chip on the Arduino Uno board. The Xloader program is used to load the hex file. When the Xloader program is executed, a window will open up, as shown in Fig. 5. Specify the COM port number for the USB socket to which the Arduino Uno board is connected. This number can be found from the 'Device Manager' on Windows on the PC/Laptop.

Browse the hex code from XLoader window as shown in Fig. 5, and press upload button. After loading the code, disconnect the Arduino board from the laptop and connect a

9V adaptor to its DC socket (or a 5V supply to its 5V and ground pins).

Construction and testing

An actual-size PCB layout for colour identification is shown in Fig. 6 and its components layout in Fig. 7. After assembling the circuit on the PCB, enclose it in a suitable box. Fix the sensor module and LCD on the front panel and connect the wires to their respective connectors on the PCB.

When you switch on power and place any coloured object in front of the sensor, the name of the colour along with its chroma value will be displayed on the LCD. Make sure that the sensor is not directly exposed to the surrounding light source to avoid false detection. **EFY**

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